St. Philomena's College (Autonomous), Mysore			
PG Department of Studies and Research in Chemistry			
Question Bank (Revised Curriculum 2020 onwards)			
First Year- First Semester (2020 -21 Batch)			
Course Title (Paper Title): FUNDAMENTALS OF CHEMICAL ANALYSIS QP Code: 85121			
Unit/ Module	**SI. No**	**Questions**	**Marks**
1	1.	Differentiate between determination and measurement in analytical chemistry.	2
1	2.	Distinguish determination and measurement in analytical chemistry.	2
1	3.	What is analysis? Write its types.	2
1	4.	Define analysis? Give its types.	2
1	5.	Explain the terms: Determination and Measurements in analytical chemistry	2
1	6.	Describe the terms: Determination and Measurements in analytical chemistry	2
1	7.	What are the factors that effects the choice of analytical methods?	2
1	8.	List out the different criteria followed in selecting an analytical method.	2
1	9.	Give the factors that effects the choice of analytical methods?	2
1	10.	Define the terms: Accuracy and Precision	2
1	11.	Describe robustness and ruggedness?	2
1	12.	Differentiate accuracy and precision	2
1	13.	Describe Sensitivity and Selectivity?	2
1	14.	Explain the terms: Accuracy and Precision	2
1	15.	What is Sensitivity and Selectivity?	2
1	16.	Explain the terms: Sensitivity and Selectivity	2
1	17.	What is Sensitivity and Selectivity?	2
1	18.	Explain the terms: robustness and ruggedness.	2
1	19.	What is robustness and ruggedness?	2
1	20.	What is absolute error and relative error?	2
1	21.	Define the terms: Absolute error and Relative error	2
1	22.	The result of an analysis was determined as 36.97g compared with the accepted value was 37.06g. What is the relative error in parts per thousand?	2
1	23.	Percentage of lead in a brass sample determined by two analysts are as follows. (i) Analyst A: 2.63, 2.60, and 2.62 (ii) Analyst B: 2.70, 2.75, and	2

		2.81. Among these two analysts who is more précised as well accurate?	
1	24.	An analyst determines value of 70.55% vitamin C in a fresh gooseberry that actually contains 70.25%. Calculate the absolute and relative error.	2
1	25.	A student reported the following percentages of FeO in the sample: 16.60, 16.74, 16.66, 16.64, 16.82 and 16.50. If the NIST reported value is 16.55%, calculate the absolute and percentage error.	2
1	26.	The result of an analysis was determined as 15.752g while the accepted value was 15.872g. Calculate the absolute and relative error.	2
1	27.	Write the factors that effects the choice of analytical methods?	2
1	28.	What are the factors that effects the choice of analytical methods?	2
1	29.	Mention the factors that effects the choice of analytical methods?	2
1	30.	Give the factors that effects the choice of analytical methods?	2
1	31.	Write normal distribution curve.	2
1	32.	How do you minimize the determinate error?	2
1	33.	Write normal distribution curve.	2
1	34.	Comment on minimization of error	2
1	35.	Distinguish between: Standard deviation and Variance	2
1	36.	Distinguish between: Average deviation and Relative average deviation	2
1	37.	What is mean and median	2
1	38.	What is mean and mean deviation.	2
1	39.	What is median and mean deviation.	2
1	40.	What is median and standard deviation.	2
1	41.	Calculate the mean and median for the results given by the replicate determination of chloride in a metal chloride: X1= 32.22 X2 = 32.64, X3 = 32.52 and X4 = 32.46.	2
1	42.	Calculate the value of 'A' if the mean of 6, 4, 7, A and 10 is 8.	2
1	43.	The mean of 40 observation was 160. On rechecking, it was found that one of the value of 165 was wrongly copied as 125 for computation of mean. Calculate the error mean.	2
1	44.	Calculate the mean of the following distribution. X= 4, 6, 9, 10, 15. f = 5, 10, 10, 7, 8.	2
1	45.	Analysis of a given quantity gave the following nine values. Assuming the errors to be random ones. Calculate mean and relative mean deviation for 46.62, 46.47, 46.64, 46.76, 46.53, 46.60, 46.71, 46.60, 46.71.	2
1	46.	What is mean and standard deviation.	2
1	47.	Define mean and median	2
1	48.	Define mean deviation and standard deviation.	2

1	49.	Define mean and mean deviation.	2
1	50.	Define median and mean deviation.	2
1	51.	Define median and standard deviation.	2
1	52.	Define mean and standard deviation.	2
1	53.	Define mean deviation and standard deviation.	2
1	54.	Write the equation for standard deviation and stand error of mean of a set of data.	2
1	55.	Write the equation for mean and standard deviation.	2
1	56.	Write the equation for mean deviation and standard deviation.	2
1	57.	Write the equation for relative mean deviation and standard deviation.	2
1	58.	Write the equation for relative standard deviation and coefficient of variance.	2
1	59.	Write the equation for standard deviation and coefficient of variance.	2
1	60.	Write the equation for relative mean deviation and coefficient of variance.	2
1	61.	Write the equation for mean and coefficient of variance.	2
1	62.	Write the equation for mean and relative mean deviation.	2
1	63.	Write the equation for average deviation and coefficient of variance.	2
1	64.	Write the equation for relative mean deviation and standard deviation.	2
1	65.	Explain the terms confidence limit and confidence interval.	2
1	66.	Write the equation used to calculate the confidence limits when a good estimate of σ is available and not available.	2
1	67.	Distinguish between confidence limits and confidence interval.	2
1	68.	Derive confidence limit and confidence interval.	2
1	69.	Give the equation used to calculate the confidence limits when a good estimate of σ is available and not available.	2
1	70.	What is F-test and t-test?	2
1	71.	A chemist analysis an iron ore for FeO and obtain a value of 12.35% with the S.D of 0.08. Calculate the 95% confidence interval of the mean based on (i) Four determinations and (ii) Eight determinations. (t tab = 3.182 when n = 4 and 2.365 when n = 8)	2
1	72.	Discuss how determinate errors are propagated when arithmetic operations like addition and subtraction performed.	2
1	73.	Explain how determinate errors are propagated when arithmetic operations like addition and subtraction performed.	2
1	74.	Explain how indeterminate errors are propagated when arithmetic operations like addition and subtraction performed.	2
1	75.	Discuss how indeterminate errors are propagated when arithmetic operations like addition and subtraction performed.	2

1	76.	Four replicate values were obtained for the determination of a pesticide in river water 0.403, 0.401, 0.380 mg/dm3. Apply the Q-test whether the last value should be retained or rejected (Q tab = 0.83)	2
1	77.	A technician obtained the following results for the determination concentration (mg/dl) of cholesterol in a blood sample: 240, 265, 230, 238 and 244 (Q tab = 0.64 at 90% confidence level) (i) Can any result be rejected by Q-test? (ii) What value should be reported for concentration?	2
1	78.	A technician obtained the following results for the determination concentration (mg/dl) of cholesterol in a blood sample: 240, 265, 230, 238 and 244 (Q tab = 0.64 at 90% confidence level) (i) Can any result be rejected by Q-test? (ii) What value should be reported for concentration?	2
1	79.	Five determinations of Vit C content of a citrous fruit drink gave the following results 0.218, 0.219, 0.230, 0.215 and 0.222 mg/ml. Apply Q test to see if the value 0.230 can be discarded (Q tab = 0.64 at 90% confidence level).	2
1	80.	Discuss how indeterminate errors are propagated when arithmetic operations like multiplication and division performed.	2
1	81.	Explain how indeterminate errors are propagated when arithmetic operations like multiplication and division performed.	2
1	82.	Describe control chart.	2
1	83.	Explain control chart.	2
1	84.	What is detection limit and linear dynamic range?	2
1	85.	Explain the terms: detection limit and linear dynamic range?	2
1	86.	What is detection limit and linear dynamic range?	2
1	87.	Describe the terms: detection limit and linear dynamic range?	2
2	88.	What is sampling?	2
2	89.	What is meant by Random sampling?	2
2	90.	Define random and Judgmental sampling.	2
2	91.	What is meant by Systematic sampling?	2
2	92.	What is meant by Judgment sampling?	2
2	93.	Write the five questions which should be consider during designing a sampling plan.	2
2	94.	Give the difference between Random and Systematic sampling.	2
2	95.	What is meant by Systematic-judgmental sampling?	2
2	96.	What is meant by stratified sampling?	2
2	97.	What is convenience sampling?	2
2	98.	Define Grab samples.	2
2	99.	What is composite sampling?	2

2	100.	Suppose you are to analyze a solid where the particles containing analyte represent only 1×10^{-7} % of the population. How many particles must be collected to give a relative sampling variance of 1%?	2
2	101.	Differentiate between Grab and composite sample.	2
2	102.	Differentiate between stratified and convenience sampling.	2
2	103.	What is meant by *in situ* sampling?	2
2	104.	If size of sample too small or too large, the sampling error will occurs. Give reasons	2
2	105.	Write the equation to describe the confidence interval for the sampling error.	2
2	106.	Write the three steps involves during implanting the sample plan	2
2	107.	What is meant by gross sample?	2
2	108.	How to implement the sampling plan if sample is homogeneous,	2
2	109.	Sample bottle for collecting solution are made up of plastic and glass. Justify.	2
2	110.	What are the disadvantages of glass sample bottle?	2
2	111.	What are the advantages of plastic sample bottle over glass bottle?	2
2	112.	What are the preservation and maximum holding time for a sample containing ammonia and chloride?	2
2	113.	What are the preservation and maximum holding time for a sample containing metals, Hg and Cr.	2
2	114.	What are the preservation and maximum holding time for a sample containing nitrate and organocholro pesticides	2
2	115.	How samples will prepare from solution during implementing of the samples?	2
2	116.	A pump is used to pull the gas into the container during implement process. Mention the advantage and disadvantage of this method.	2
2	117.	Difficulty of storing gas samples. Justify	2
2	118.	What is organic polymeric sorbent?	2
2	119.	What is meant by breakthrough volume?	2
2	120.	Name the process to bring solid samples into solution.	2
2	121.	How a solid sample is preserved?	2
2	122.	How to preserve and prepare a Gas sample?	2
2	123.	Why standardizing of titrant is required?	2
2	124.	How titrant is selected for neutralization reaction?	2
2	125.	Solid NaOH is contaminated with carbonate. Justify	2

2	126.	How to prepare solution of carbonate free- sodium hydroxide?	2
2	127.	How CO_2 acts corroding agent in water?	2
2	128.	Write the two procedure to analysis of nitrogen in compound	2
2	129.	How to determine the element sulphur in the compounds	2
2	130.	Write the five titration for combination of sodium hydroxide, sodium bicarbonate and sodium carbonate.	2
2	131.	Solution of same amount of bicarbonate and hydroxide ions can't exist in nature. Give reasons.	2
2	132.	Write the steps involves in Kjeldahl's method.	2
2	133.	Write the two ways to determine concentration of ammonia in solution.	2
2	134.	What is an equivalent mass? Give an example.	2
2	135.	Define equilibrium constant with an example.	2
2	136.	Calculate the equivalent weight for $KMnO_4$ and $K_2Cr_2O_7$	2
2	137.	Calculate the equivalent weight for $KMnO_4$ and H_2SO_4	2
2	138.	Many carboxylic acid are insoluble in water. How to analysis like this acids using titration method.	2
2	139.	How to determine equivalent weight of an acid?	2
2	140.	How to determine equivalent weight of a base?	2
2	141.	Why acid base titrimetry play an important role in quantitative analysis?	2
2	142.	What is the principal of titrimetric analysis?	2
2	143.	What is role of non-aqueous solvents in acid-base titration?	2
2	144.	Why we used some time non aqueous solvents in acid-base titration?	
2	145.	What are the problems arises after using non aqueous solvents in acid-base titration?	2
2	146.	What is meant by amphiprotic solvent?	2
2	147.	Give the difference between acidic and basic solvents.	2
2	148.	Differentiate between neutral and inert solvents. Give an example	2
2	149.	What is meant by titrants? Give example	2
2	150.	What is meant by standards? Give example	2
2	151.	Define acidic titrants with an example.	2
2	152.	Name any two standards used for acidic titrants.	2

2	153.	Give any two example of basic titants.	2
2	154.	Differentiate acidic and basic titrants.	2
2	155.	What is meant by equivalence point?	2
3	156.	Write the chemical structure of phenolphthalein in acid and base form	2
3	157.	How do you estimate the atmospheric sulphur di oxide using acid base titrations? Write its principle	2
3	158.	Why thiocyanate is added in volhard method for the determination of chloride. Mention the indicator used in this method	2
3	159.	Discuss any two applications of redox titration	2
3	160.	What is displacement titration? Give an example	2
3	161.	Propose a scheme for determination of nitrogen in an ammonium salt by acid-basetitrimetry	2
3	162.	What is effect of concentration on pH at the equivalence point in an acid-base titration curve?	2
3	163.	Explain the Fajan's method of precipitation titration	2
3	164.	How is COD determined in waste water?	2
3	165.	Why standardization of titrant is required? How titrant is selected for neutralization titration	2
3	166.	Write the structure of EBT indicator. Explain the action of indicator in complexometric titration	2
3	167.	What are acid-base indicators? Write the indicator action of phenolphthalein	2
3	168.	Write the indicator action of potassoim chromate in mohr method of titrating chloride with Ag (I)	2
3	169.	Write the indicator action of EBT in the titration of Zn (II) with EDTA at pH 10	2
3	170.	Explain the terms equivalence point and end point	2
3	171.	What are precipitation titration? Why their scope is limited	2
3	172.	Why phosphoric acid is used in the titration of Fe (II) with dichromate solution	2
3	173.	Write the structures of EBT and Patton and Reeders's indicators	2
3	174.	Why is nitrobenzene is added to the solution in the titration of chloride by volhard method	2
3	175.	What is the pH at the equivalence point for the titration of a weak acid against weak base? Why?	2
3	176.	What criterion is used in selecting a redox indicator for a particular redox titration?	2
3	177.	Give an example of the use of masking agent	2
3	178.	Titration of 0.485 g sample by Mohr method required 36.8 ml of 0.106 M silver nitrate solution. Calculate the percentage of chloride in the sample	2
3	179.	Give two circumstances in which an EDTA back titration might be necessary	2

3	180.	Explain with a example each the auxillary reducing and oxidizing agents	2
3	181.	What is masking agent: Give one example of the masking agent in the titration of Mg (II) in the presence of Cd (II) and Zn (II)	2
3	182.	How nitrates and nitrites are determined by neutralization titration	2
3	183.	Write the indicator action of potassium chromate in mohr's method of titrating chloride with Ag (I)	2
3	184.	What is titration curve? Explain with an example	2
3	185.	Explain how the end point is detected by monitoring temperature.	2
3	186.	Calculate the equivalence point of 10 ml of solution of 0.1 M sulphuric acid being titrated against 0.1 M NaOH	2
3	187.	Calculate the pH of 0.1 M acetic acid (given that $K_a = 1.75 \times 10^{-5}$)	2
3	188.	Why standardization of a titrant like $KMnO_4$ is carried out against sodium oxalate.	2
3	189.	List the factors affecting the shape of the titration curve in precipitation titration	2
3	190.	What is ferroinindicator? Write its structure	2
3	191.	Titration of 0.246 g of dolomite by complexometric titration consumed 20 mL of 0.1 M EDTA. Calculate the amount of calcium in the given sample	2
3	192.	What are metallochromicindicators? Explain the indicator action of Eriochrome black T in the titration of Mg (II) ion with EDTA at pH with its structure	2
3	193.	A 50.00-mL sample of a citrus drink requires 17.62 mL of 0.04166 M NaOH to reach the phenolphthalein end point. Express the sample's acidity in terms of grams of citric acid, $C_6H_8O_7$, per 100 mL.	2
3	194.	The concentration of Cl– in a 100.0-mL sample of water drawn from a fresh water aquifer suffering from encroachment of sea water, was determined by titrating with 0.0516 M $Hg(NO_3)_2$. The sample was acidified and titrated to the diphenylcarbazone end point, requiring 6.18 mL of the titrant. Report the concentration of Cl– in parts per million.	2
3	195.	A 250.0-mg sample of an organic weak acid was dissolved in an appropriate solvent and titrated with 0.0556 M NaOH, requiring 32.58 mL to reach the end point. Determine the compound's equivalent weight.	2
3	196.	An indication of the average formula weight of a fat is its saponification number, expressed as milligrams of KOH required to hydrolyze (saponify) 1 g of fat. A 1.10 g sample butter is treated with 25.0 ml of 0.250 M KOH solution. After saponification is complete, the unreacted KOH is back titrated with 0.250M HCl, requiring 9.26 ml. What is the saponification number of the fat.	2
3	197.	A sample containing amino acid residue, $CH_3CH(NH_2)COOH$, plus inert matter analyzed by the Kjeldahl method. A 2.00 g sample is digested, the NH_3 distilled and collected in 50.0 ml of 0.150 M sulphuric acid, and a volume of 9.0 ml of 0.100 M NaOH is required for back titration. Calculate the percent alanine in the sample.	2
3	198.	An EDTA solution is standardized against high purity $CaCO_3$ by dissolving	2

		0.3982 g $CaCO_3$ in hydrochloric acid, adjusting pH to 10 with ammonicalbuffer, and titrating. If, 38.26 ml was required for the titration, what is the molarity of EDTA?	
3	199.	Calcium in powdered milk is determined by ashing a 1.5 g sample and then titrating the calcium with EDTA solution, 12.1 ml being required. The EDTA was standardized by titrating 10.0 ml of a Zinc solution prepared by dissolving 0.632 g Zinc metal in acid and diluting to 1 L (10.8 ml EDTA required for titration). What is the concentration of calcium in the powdered milk in parts per million?	2
3	200.	A 523.1 mg sample of impure KBr is treated with excess silver nitrate and 814.5 mg AgBr is obtained. What is the purity of the KBr?	2
3	201.	Chloride in brine solution is determined by Volhard method. A 10.00 ml aliquot of the solution is treated with 15.00 ml of standard 0.1182 M silvernitrate solution. The excess silver is titrated with standard 0.101 M KSCN solution, requiring 2.38 ml to reach the red $Fe(SCN)^{2+}$ end point. Calculate the concentration of chloride in the brine solution, in g/L	2
3	202.	A 0.200 g sample containing copper is analysediodometrically, copper(II) is reduced to Copper(I) by iodide. What is the percent copper in the sample if 20.0 ml of 0.100 M sodium thiosulphate is required for titration of the liberated iodine?	2
3	203.	A solution of sodiumthiosulphate is standardized iodometrically against 0.1262 g of high purity $KBrO_3$ (molecular mass of $KBrO_3$=167.01), requiring 44.97 ml sodium thiosulphate. What is the molarity of sodium thiosulphate?	2
3	204.	A 0.200 g sample containing copper is analysediodometrically. Copper (II) is reduced to Copper (I) by iodide. What is the percent copper in the sample of 20.0 mL of 0.100 M sodium thiosulphate is required for titration of the liberated iodine?	2
3	205.	A solution of sodium thiosulphate is standardized iodometrically against 0.1262 g of high purity $KBrO_3$, requiring 44.97 mL sodium thiosulphate. What is the molarity of the sodium thiosulphate?	2
3	206.	The calcium in a 5.00 mL serum sample is precipitated as calcium oxalate with ammonium oxalate. The filtered precipitate is dissolved in acid, the solution is heated, and the oxalate is titrated with 0.001 M potassium permanganate, requiring 4.94 mL. calculate the concentration of calcium in the serum in meq/L	2
3	207.	Consider the titration of Sn (II) with Fe (III) in HCl medium. Using Pt and calomel electrodes. Write a balanced titration reaction, two half reactions for the indicator electrode and two Nernst equations for the cell voltage	2
3	208.	A sample of ground water contains both calcium and magnesium hardness. How they are determined by EDTA titration?	2
3	209.	A solution contains both carbonate and bicarbonate. How do you propose to detect and determine their concentrations by titrating the solution with HCl using phenolphthalein and methyl red indicators	2
3	210.	With relevant chemistry propose schemes for the determination of elemental sulphur and nitrate nitrogen by acid-basetitrimetry	2
3	211.	With relevant chemistry, propose schemes for the determination of nitrogen in an ammonium salt and sulphur di oxide in a sample of air by acid-basetitrimetry	2

3	212.	Describe the general methods for performing EDTA titrations. What are the advantages of each	2
3	213.	Explain the basic principle and method for determination of the carbonates and bicarbonates in the given sample	2
3	214.	What is alkalinity of a water sample? Mention the sources of alkalinity of water sample and explain the determination of alkalinity of water sample	2
3	215.	What is precipitation titration? Describe different types of precipitation titration with principles and indicators involved in them	2
1	216.	Give the classification of analytical methods	4
1	217.	Write the classification of analytical methods	4
1	218.	Explain the classification of analytical techniques.	4
1	219.	Give the classification of analytical techniques.	4
1	220.	Define error. Write the ways of determining errors with equation.	4
1	221.	What is an error? Write the ways of determining errors with equation.	4
1	222.	What is an error? Calculate the absolute error, relative error and percentage error of the following data: 19.2, 19.5, 19.3, 20.0, 19.7, 19.7 and the true value is 19.6	4
1	223.	What is an error? Write the types and sources of errors.	4
1	224.	Define error. Explain the types and sources of errors.	4
1	225.	Write the salient features of determinate and indeterminate errors.	4
1	226.	Write a note on determinate and indeterminate errors.	4
1	227.	Explain determinate and indeterminate errors.	4
1	228.	Classify and explain different types of errors observed in chemical analysis.	4
1	229.	Outline determinate and indeterminate errors.	4
1	230.	Draw the normal error curve and write its properties.	4
1	231.	Explain the methods of minimizing determinate errors.	4
1	232.	Draw and Explain the normal error curve. Discuss the salient features of this curve.	4
1	233.	How do you minimize the determinate errors?	4
1	234.	Analysis of a sample of silver alloy gave the following percentage values for the silver content: 7.08, 7.21, 7.12, 7.09, 7.14, 7.07, 7.14, 7.18, and 7.11. Calculate the mean, standard deviation and coefficient of variance for the set of data.	4
1	235.	Analysis of a sample of haematite gave the following percentage values for the iron content: 7.08, 7.21, 7.12, 7.09, 7.14, 7.07, 7.14, 7.18, 7.11. Find out the mean, standard deviation and coefficient of variance for the values.	4

1	236.	Explain mean and median	4
1	237.	Explain mean and mean deviation.	4
1	238.	Explain median and mean deviation.	4
1	239.	An analysis to determine the percentage weight of copper in an ore sample gives the following results: 16.45%, 16.30%, 16.64%, 16.67%, 16.70% and 16.49% (w/w). Calculate the mean, standard deviation and range for this data set.	4
1	240.	Explain median and standard deviation.	4
1	241.	Discuss the various types of determinate errors encountered in analysis and how are they minimized?	4
1	242.	Explain mean and standard deviation.	4
1	243.	The following percentage of FeO were recorded in a sample: 5.683, 5.620, 5.551, 5.549, 5.536, 5.552, 5.548, 5.539, 5.554, 5.552, 5.684, and 5.632. Report the mean, median, range, standard deviation and variance for these data.	4
1	244.	What is F-test? Write the steps involved in applying the test.	4
1	245.	What is t-test? Write the steps involved in applying the test.	4
1	246.	What is t-test? Outline the steps involved in applying the test.	4
1	247.	Write a note on student's t-test.	4
1	248.	Write a note on F-test.	4
1	249.	A blood sample is sent to two laboratories to be analyzed for cholesterol. The results obtained for the concentration (mg/dl) were (i) Lab 1: Mean = 243, S1 = 13, n1 = 5 (ii) Lab 2: Mean = 258, S2 = 15, n2 = 6. (1) Are the standard deviation significantly different at the 95% level? (Ftab = 6.26). (2) Are the two means significantly different at the same confidence level? (t tab = 2.306)	4
1	250.	Two methods for the determination of polyaromatic hydrocarbons in soils were compared by analyzing a standard with the following results. Number of determination by each method: 10, Nubmer of degrees of freedom: 18, UV Spectrophotometry: mean = 28 mg/kg ; S = 0.3mg/kg. Fluorimetry: mean = 26.25 mg/kg ; S = 0.23 mg/kg. Do the mean for the two methods differ significantly (t tab = 2.10 at 95% confidence level)	4
1	251.	Standard sample of iron in the analysis gives the following data: 16.65, 16.68, 16.60, 16.70, 16.58 and 16.63. S.D obtained from standard method is 0.062, estimate the range, median and standard deviation. Find the variance in the given analysis.	4
1	252.	A sample of soda ash is analyzed by two different methods, giving the following results for the percentage of Na2CO3. **Method 1**: Mean % is 42.34, SD is 0.10, Number of analysis are 5. **Method 2:** Mean % is 42.44, S.D is 0.12, Number of analysis are 4. (i) Are the S.D significantly different? (F= 6.59) (ii) Are the two means significantly different at 95% probability level? (t = 2.365)	4
1	253.	Explain the significance of student's t-test and F- test in treating experimental	4

		data.	
1	254.	A technician obtained the following results for the determination concentration (mg/dl) of cholesterol in a blood sample: 240, 265, 230, 238 and 244. (Qtab = 0.64 at the 90% confidence level). 1) Can any result be rejected by Q-test? 2) What value should be reported for concentration?	4
1	255.	Discuss how determinate errors are propagated when arithmetic operations like multiplication and division performed.	4
1	256.	Explain how determinate errors are propagated when arithmetic operations like multiplication and division performed.	4
1	257.	Explain how indeterminate errors are propagated when arithmetic operations like addition, subtraction and multiplication, division are performed.	4
1	258.	Discuss how indeterminate errors are propagated when arithmetic operations like addition, subtraction and multiplication, division are performed.	4
1	259.	What is Q-test? Write the steps involved in applying the test.	4
1	260.	Write a note on Q-test.	4
1	261.	What is Q-test? Write the steps involved in applying the test.	4
1	262.	The analysis of calcite samoleyieldingCaO percentage of 55.95, 56.00, 56.04, 56.08 and 56.23. the last value appears anomalous. Should it be rejected or retained? Q tab at 90% confidence level is 0.54	4
1	263.	What are determinate errors? Discuss on some common determinate errors.	4
1	264.	Describe the method of least squares for driving calibration curves.	4
1	265.	Write a note on method of least squares for driving calibration curves.	4
1	266.	Describe control chart.	4
1	267.	Explain control chart.	4
1	268.	Write a note on control chart.	4
1	269.	Write a note on comparison of two means and two standard deviations.	4
1	270.	Explain comparison of two means and two standard deviations.	4
1	271.	Describe comparison of two means and two standard deviations.	4
1	272.	Give the classification of analytical methods	5
1	273.	Write the classification of analytical methods	5
1	274.	Explain briefly the classification of analytical techniques.	5
1	275.	What is an error? Write the types and sources of errors.	5
1	276.	Define error. Explain the types and sources of errors.	5
1	277.	What is an error? Write the salient features of determinate and indeterminate errors.	5

1	278.	Write a note on determinate and indeterminate errors.	5
1	279.	Define error? Explain determinate and indeterminate errors.	5
1	280.	What is an error? Outline determinate and indeterminate errors.	5
1	281.	What is an error? Classify and explain different types of errors observed in chemical analysis.	5
1	282.	What is an error? Classify and explain different types of errors observed in chemical analysis.	5
1	283.	What is determinate error? What is the source of determinate error? How it is minimized?	5
1	284.	An analyst reported the following percentage of FeO in a sample, 16.65, 16.70, 16.58, 16.60, 16.38 and 16.63. From this set of results calculate mean, median, average deviation, relative average deviation and coefficient variation.	5
1	285.	Classify and explain different types and sources of errors observed in chemical analysis.	5
1	286.	Explain different types and sources of errors observed in chemical analysis.	5
1	287.	Explain Gaussian distribution curve. How do you minimize the determinate error?	5
1	288.	Analysis of a sample of silver alloy gave the following percentage values for the silver content: 7.08, 7.21, 7.12, 7.09, 7.14, 7.07, 7.14, 7.18, and 7.11. Calculate the mean, median, standard deviation and coefficient of variance for the set of data.	5
1	289.	Explain the various types of determinate errors encountered in analysis and how are they minimized?	5
1	290.	An analysis to determine the percentage weight of copper in an ore sample gives the following results: 16.45%, 16.30%, 16.64%, 16.67%, 16.70% and 16.49% (w/w). Calculate the mean, median, standard deviation and range for this data set.	5
1	291.	What is F-test? Write the steps involved in applying the test.	5
1	292.	What is t-test? Write the steps involved in applying the test.	5
1	293.	What is Q-test? Write the steps involved in applying the test.	5
1	294.	A Chemist developed a new method for the analysis of iron in a sample and obtained the following results: 18.64mg; 19.84mg; 20.45 mg; 20.56 mg; and 19.29 mg. The standard method was also used to analyze the sample and obtained the following results: 20.54 mg; 20.83 mg; 20.42 mg; 20.68 mg. Find whether the new method is having significant difference from the standard method at 95% confidence level. (F table = 6.59)	5
1	295.	A blood sample is sent to two laboratories to be analyzed for cholesterol. The results obtained for the concentration (mg/dl) were (i) Lab 1: X1 = 243, S1 = 13, n1 = 5 (ii) Lab 2: X2 = 258, S2 = 15, n2 = 6. (1) Are the standard deviation significantly different at the 95% level? (Ftab = 6.26). (2) Are the two means significantly different at the same confidence level? (t tab = 2.306)	5
1	296.	Standard sample of iron in the analysis gives the following data: 16.65, 16.68, 16.60, 16.70, 16.58 and 16.63. S.D obtained from standard method is 0.062,	5

		estimate the range, median and standard deviation. Find the variance in the given analysis.	
1	297.	A new high performance liquid chromatographic method for the determination of Pseudoephedrine in a pharmaceutical product at two different levels compared with an established method with the following results. (i) **Method 1**: 59.9, 59.3, 60.4, 59.6, 60.2, 60.7 (ii) **Method 2**: 58.6, 58.3, 60.5, 60.7, 59.2, 59.8. Do the means of two methods differ significantly at the confidence level? (t tab = 2.228)	5
1	298.	A blood sample is sent to two laboratories to be analyzed for cholesterol. The results obtained for the concentration (mg/lit) were (i) Lab 1: Mean = 243, S1 = 0.13, n1 = 11 (ii) Lab 2: Mean = 248, S2 = 0.15, n2 = 11. (1) Are the standard deviation significantly different at the 95% level? (2) Are the mean significantly different at the 95% confidence level? (Ftab = 2.98 and t tab = 2.086)	5
1	299.	What is t-test? Indicate its significance in determining the reliability of experimental results. Outline the steps involved in applying the test.	5
1	300.	What is t-test? Explain the steps involved in applying t-test.	5
1	301.	How determinate errors are propagated when arithmetic operations like addition, subtraction and multiplication, division performed?	5
1	302.	Write a note on propagation of determinate errors when arithmetic operations like addition, subtraction and multiplication, division performed.	5
1	303.	A technician obtained the following results for the determination concentration (mg/dl) of cholesterol in a blood sample: 240, 265, 230, 238 and 244. (Qtab = 0.64 at the 90% confidence level). 1) Can any result be rejected by Q-test? 2) What value should be reported for concentration?	5
1	304.	What are determinate errors? Discuss on some common determinate errors.	5
1	305.	Explain the method of least squares for driving calibration curves.	5
1	306.	Outline the method of least squares for driving calibration curves.	5
1	307.	Briefly explain the least squares method to obtain the best straight linen through the set of data points for a Calibration Curve.	5
1	308.	Comment on the least square method of driving calibration curves.	5
1	309.	Elaborate control chart.	5
1	310.	Describe control chart.	5
1	311.	Explain control chart.	5
1	312.	Write a note on control chart.	5
1	313.	Write a note on comparison of two means and two standard deviations.	5
1	314.	Explain comparison of two means and two standard deviations.	5
1	315.	Describe comparison of two means and two standard deviations.	5
2	316.	What is sampling? Explain the random and Judgment sampling.	5
2	317.	Summarize a report on Random, Judgmental and Systematic-judgmental	5

		sampling.	
2	318.	Explain the importance of sampling of sampling.	5
2	319.	Write a note on designing a sample plan.	5
2	320.	What is sampling? Discuss random and systematic sampling.	5
2	321.	Write a note on i) Random sampling ii) Systematic sampling	5
2	322.	Summarize a report on the number of samples to collect for sampling.	5
2	323.	Write a note on size of sample for sampling.	5
2	324.	Explain how much sample to collect for the sampling.	5
2	325.	Write a note on i) Stratified and convenience sampling ii) Grab and composite Sample	5
2	326.	Discuss Stratified and convenience sampling in detail.	5
2	327.	Summarize a report on Grab, composite and *in situ* sampling.	5
2	328.	In an analysis for the inorganic ash content of a breakfast cereal required a sample of 1.5 g to establish a relative standard deviation for sampling of ±2.0%. How many samples are needed to obtain a relative sampling error of no more than 0.80% at the 95% confidence level?	5
2	329.	Explain implementing the sampling plan for a solutions sample.	5
2	330.	Summarize a report on implementation of sampling plan for a solution.	5
2	331.	Write a note on sample collection, preservation and preparation for solutions sample implantation.	5
2	332.	Discuss the sample preservation of solutions sample during implementing the sample plan.	5
2	333.	Explain implementing the sampling plan for Gases.	5
2	334.	Summarize a report on implementation of sampling plan for Gases sample.	5
2	335.	Write a note on sample collection, preservation and preparation for Gases sample implantation.	5
2	336.	Explain implementing the sampling plan for solids.	5
2	337.	Summarize a report on implementation of sampling plan for solids.	5
2	338.	Write a note on sample collection, preservation and preparation for solids sample implantation.	5

2	339.	Explain the process digestion and decomposition process for bringing solid sample into solutions.	5
2	340.	Write a note on bringing solid samples into solution.	5
2	341.	Explain the analysis of free CO_2 in water. Write the equation involves in it.	5
2	342.	Write a note on analysis of free CO_2 in water. Write the equation involves in it.	5
2	343.	Discuss the inorganic analysis like acidity and alkalinity.	5
2	344.	How alkalinity of solution will determine? Explain with example	5
2	345.	Summarize a report on the analysis of free CO_2 in water and waste water.	5
2	346.	Explain the Kjeldahl's method for analysis of nitrogen.	5
2	347.	Write a note on analysis of carbonate and bicarbonates	5
2	348.	Summarize a report on analysis of ammonium salts, nitrates and nitrites.	5
2	349.	Discuss the determination/analysis of nitrogen and sulphur elements.	5
2	350.	Write a note on the analysis of i) Carbonate and bicarbonate ii) Nitrate and nitrites	5
2	351.	What is an equivalent weight? Discuss the analysis of functional group like carboxylic and sulphonic acids.	5
2	352.	Summarize a report on determination of amine and ester group.	5
2	353.	Explain the determination of Hydroxyl group and carbonyl group.	5
2	354.	Write a note the Air pollutant like SO_2 and its determination.	5
2	355.	What is an equilibrium constant? Hence, explain the how to determine the equivalent weight of acid, base and salt.	5
2	356.	Explain the role of solvent in acid base titration.	5
2	357.	Write a note on the differentiating ability of a solvent.	5
2	358.	Summarize a report on types of solvent system.	5
2	359.	Write a note on leveling effect.	5
2	360.	Discuss the properties of some selected solvent.	5
2	361.	Write a note on important characteristic of amphiprotic solvents.	5
2	362.	Summarize a report on autoprotolysis.	5
2	363.	Write a note on titrants and standards used in acid base titration.	5
2	364.	Explain the effect of water in titrimetric analysis.	5
2	365.	Summarize a report on determining the equivalence point.	5

2	366.	Discuss the titrimetric analysis in the determination of carboxylic acid, phenols and amine.	5
2	367.	Write a note on titration curves in acid base titration.	5
3	368.	A 50.00 ml of aliquot of 0.100 M NaCl solution is titrated with 0.100 M silver nitrate. Calculate the pCl values after adding 0.0, 10.0, 49.9, 50.0, 60.0 mL of the titrant. Sketch the titration curve (given $K_{sp} = 1.8\times 10^{-10}$)	5
3	369.	Discuss the theory of redox titration with an example	5
3	370.	Explain the difference between absolute stability constant and conditional stability constant. How they are related. Add a note on the importance of conditional stability constant in EDTA titration.	5
3	371.	What are precipitation titrations? Why their scope limited? Write the mechanism of action of fluorescein in the titration of chloride with Ag(I)	5
3	372.	Explain the differences between Mohr and volhard method for the determination of chloride. Indicate the advantages and limitations of both methods	5
3	373.	Sulphate can be determined indirectly by precipitating $BaSO_4$ with excess Ba^{2+} and titrating the excess with standard EDTA. 0.3260 g sample containing sulphate is dissolved, and the sulphate is precipitated by adding 25.00 mL of 0.03120 M $BaCl_2$. After the removal of $BaSO_4$ by filtration, the excess of Ba^{2+} is titrated with 26.38 mL of 0.01822 M EDTA. Calculate the percentage of sulphate in the sample.	5
3	374.	Illustrate the effect concentration of titrant and analyte, dissociation constant of weak acid on the shape of titration curve involving weak acid and strong base	5
3	375.	For the titration of 25 mL of 0.025 M NaOH solution with 0.025 M HCl. Calculate the pH after 0.0, 10.0, 15.0, 20.0, 25.0, 30.0 ml of titrant being added. Sketch the titration curve	5
3	376.	Write a note on the following a) determination of alkalinity of waste water by neutralization titration b) determination carboxylic acid and amines	5
3	377.	A 50.0 ml of 0.005 M $Ca^{2+}$was titrated with 0.01 M EDTA at pH 10. Calculate the pCa values at the start of titration, after the addition 15.0, 20.0, 25.0, 30.0 ml of titrant. (Given $K_{abs}= 5 \times 10^{10}$ and $\alpha_{(y^{4-})}= 0.35$ at pH 10)	5
3	378.	What are precipitation titrations? Write the importance of K_{sp} and application of precipitation titration for the determination of chloride content by Mohr's method	5
3	379.	Calculate pAg of the solution during the titration of 50.0 ml of 0.1 M NaCl with 0.1 M $AgNO_3$ after 0.0, 10.0, 30.0, 50.0, 60.0 ml of titrant added (K_{sp} of AgCl= 1.82×10^{-10})	5
3	380.	Write a note on finding end point with visual indicator and by potentiometry	5
3	381.	Sketch the titration curve for 50.0 ml of 0.1 M Fe (II) and 0.1 M Ce(IV) in matrix of 1 M $HClO_4$ at 10.0, 20.0, 30.0, 40.0, 50.0, 60.0, 70.0, 80.0 ml of Ce (IV) (standard redox potential of Fe (III)/Fe (II) = 0.761 V and Ce (IV)/Ce (III) = 1.70 V)	5
3	382.	Calculate the pH during the titration of 50.00 ml of 0.050 M NaOH with 0.100 M HCl after the addition of the following volume of the titrant 24.50, 25.0 and 25.50 ml. Plot the titration curve and suggest suitable indicator	5

3	383.	Derive the equation for absolute stability constant of a complex ion. How is related to the conditional stability constant? Explain the importance of conditional stability constant in EDTA titration	5
3	384.	What is COD if a sewage? Explain the method for determining the COD of sewage sample	5
3	385.	A weak acid (HA), pKa= 5.0 was titrated with 1.00 M KOH. The acid solution had a volume of 100.00 ml and a molarity of 0.100 M. Find the pH at V_b = 0.0, 5.0, 10.0, 12.0 and sketch the titration curve	5
3	386.	Explain, with suitable examples, the peroxidation and prereduction of analyte sample in redox titration. Why is it important to be able to destroy the reagents used for these purposes?	5
3	387.	Outline the procedure with relevant chemistry for the determination of total chlorine residual in public water suppliers. Explain the effects of oxidizing and reducing interferents on the result of chlorine analysis	5
3	388.	Compare Mohr method and Volhard method for the determination of chloride with respect to methodology and indicator employed. List out the advantages and disadvantages of each method	5
3	389.	Propose schemes for the determination of following by acid base titrimetry a) SO_2 in air b) Nitrogen in ammonium salt c) Ester group in organic compounds d) carbonyl group in organic compounds e) alcohol group of organic compounds	5
3	390.	With a relevant chemistry explain the following process a) COD of waste water b) Titration of organic compounds with periodate c) DO in polluted water	5
3	391.	The purity of a pharmaceutical preparation of sulfanilamide, $C_6H_4N_2O_2S$, can be determined by oxidizing the sulfur to SO_2 and bubbling the SO_2 through H_2O_2 to produce H_2SO_4. The acid is then titrated with a standard solution of NaOH to the bromothymol blue end point, where both of sulfuric acid's acidic protons have been neutralized. Calculate the purity of the preparation, given that a 0.5136-g sample required 48.13 mL of 0.1251 M NaOH.	5
3	392.	The amount of protein in a sample of cheese is determined by a Kjeldahl analysis for nitrogen. After digesting a 0.9814-g sample of cheese, the nitrogen is oxidized to NH_4^+, converted to NH_3 with NaOH, and distilled into a collection flask containing 50.00 mL of 0.1047 M HCl. The excess HCl is then back titrated with 0.1183 M NaOH, requiring 22.84 mL to reach the bromothymol blue end point. Report the %w/w protein in the cheese given that there is 6.38 g of protein for every gram of nitrogen in most dairy products.	5
3	393.	An alloy of chromel containing Ni, Fe, and Cr was analyzed by a complexation titration using EDTA as the titrant. A 0.7176-g sample of the alloy was dissolved in HNO_3 and diluted to 250 mL in a volumetric flask. A50.00-mL aliquot of the sample, treated with pyrophosphate to mask the Fe and Cr, required 26.14 mL of 0.05831 M EDTA to reach the murexide end point. A second 50.00-mL aliquot was treated with hexamethylenetetramine to mask the Cr. Titrating with 0.05831 M EDTA required 35.43 mL to reach the murexide end point. Finally, a third 50.00-mL aliquot was treated with 50.00 mL of 0.05831 M EDTA, and back titrated to the murexide end point with 6.21 mL of 0.06316 M Cu^{2+}. Report the weight percents of Ni, Fe, and	5

		Cr in the alloy.	
3	394.	The amount of Fe in a 0.4891-g sample of an ore was determined by a redox titration with $K_2Cr_2O_7$. The sample was dissolved in HCl and the iron brought into the +2 oxidation state using a Jones reductor. Titration to the diphenylamine sulfonic acid end point required 36.92 mL of 0.02153 M $K_2Cr_2O_7$. Report the iron content of the ore as %w/w Fe_2O_3**.**	5
3	395.	A 25.00-mL sample of a liquid bleach was diluted to 1000 mL in a volumetric flask. A 25-mL portion of the diluted sample was transferred by pipet into an Erlenmeyer flask and treated with excess KI, oxidizing the OCl– to Cl–, and producing I_3^-. The liberated I_3^- was determined by titrating with 0.09892 M Na2S2O3, requiring 8.96 mL to reach the starch indicator end point. Report the %w/v NaOCl in the sample of bleach.	5
3	396.	The amount of ascorbic acid, $C_6H_8O_6$, in orange juice was determined by oxidizing the ascorbic acid to dehydroascorbic acid, $C_6H_6O_6$, with a known excess of I_3^-, and back titrating the excess I_3^- with $Na_2S_2O_3$. A 5.00-mL sample of filtered orange juice was treated with 50.00 mL of excess 0.01023 M I_3^-. After the oxidation was complete, 13.82 mL of 0.07203 M $Na_2S_2O_3$ was needed to reach the starch indicator end point. Report the concentration of ascorbic acid in milligrams per 100 mL.	5
3	397.	A mixture containing only KCl and NaBr is analyzed by the Mohr method. A 0.3172-g sample is dissolved in 50 mL of water and titrated to the Ag_2CrO_4 end point, requiring 36.85 mL of 0.1120 M $AgNO_3$. A blank titration requires 0.71 mL of titrant to reach the same end point. Report the %w/w KCl and NaBr in the sample.	5
3	398.	The protein in a 1.2846-g sample of an oat cereal is determined by the Kjeldahl procedure for organic nitrogen. The sample is digested with H_2SO_4, the resulting solution made basic with NaOH, and the NH_3 distilled into 50.00 mL of 0.09552 M HCl. The excess HCl is then back titrated using 37.84 mL of 0.05992 M NaOH. Given that the protein in grains averages 17.54% w/w N, report the %w/w protein in the sample of cereal**.(**	5
3	399.	The concentration of CO_2 in air can be determined by an indirect acid–base titration. A sample of the air is bubbled through a solution containing an excess of $Ba(OH)_2$, precipitating $BaCO_3$. The excess $Ba(OH)_2$ is back titrated with HCl. In a typical analysis, a 3.5-L sample of air was bubbled through 50.00 mL of 0.0200 M $Ba(OH)_2$. Back titrating with 0.0316 M HCl requires 38.58 mL to reach the end point. Determine the parts per million of CO_2 in the sample of air, given that the density of CO_2 at the temperature of the sample is 1.98 g/L.	5
3	400.	The purity of a synthetic preparation of methylethyl ketone (C_4H_8O) can be determined by reacting the ketone with hydroxylamine hydrochloride, liberating HCl. In a typical analysis, a 3.00-mL sample was diluted to 50.00 mL and treated with an excess of hydroxylamine hydrochloride. The liberated HCl was titrated with 0.9989 M NaOH, requiring 32.68 mL to reach the end point. Report the percent purity of the sample, given that the density of methylethyl ketone is 0.805 g/mL.	5
3	401.	A 2.00 ml serum sample is analysed for protein by the modified Kjeldahl method. The sample is digested, the ammonia is distilled into boric acid solution, and 15.0 ml of the standard HCl is required for the titration of the	5

		ammonium borate. The HCl is standardized by treating 0.330 g pure ammoniumsulphate in the same manner. If 33.3 ml acid is required in the standardization titration, what is the concentration of protein in the serum in g % (Wt/Vol)?	
3	402.	The calcium in a 5.00 ml serum sample is precipitated as CaC_2O_4 with ammonium oxalate. The filtered precipitate is dissolved in acid, the solution is heated and the oxalate is titrated with 0.001 M $KMnO_4$, requiring 4.94 ml. Calculate the concentration of calcium in the serum sample in meq/L	5
3	403.	The purity of hydrazine, N_2H_4 sample is determined by titration with iodine. A sample of the oily liquid weighing 1.4286 g is dissolved in water and diluted to 1L in a volumetric flask. A 50.0 mL aliquot is taken with a pipette and titrated with standard iodine solution, requiring 42.41 mL.The iodine was standardized against 0.4123 g primary standard As_2O_3 by dissolving it in a small amount of NaOH solution, adjusting the pH to 8.0 and titrating, requiring 40.28 mL iodine solution. What is the percent purity by weight of the hydrazine?	5
3	404.	A 2.50 g of sample containing As_2O_5. Na_2HAsO_3 and inert material is dissolved and the pH is adjusted to neutral with excess $NaHCO_3$. The As (III) is titrated with 0.150 M iodine solution, requiring 11.3 mL to just reach end point. The solution (all the arsenic in the +5 state now) is acidified with HCl, excess KI is added, and the liberated iodine is titrated with 0.120 M sodium thiosulphate, requiring 41.2 mL. Calculate the percent As_2O_5 and Na_2HAsO_3 in the sample.	5
3	405.	Selenium in a 10.0 g soil sample is distilled as the tetrabromide, which is collected in aqueous solution where it is hydrolysed to SeO_3^{2-}. The SeO_3^{2-} is determined iodometrically, requiring 4.5 mL of standard thiosulphate solution for titration. If the thiosulphate titer is 0.049 mg potassium dichromate/mL.What is the concentration of selenium in the soil in ppm?	5
3	406.	A 10.00 ml of aliquot of 0.100 M NaOH solution is titrated with 0.100 M HCl. Calculate the pH values after adding 0.0, 2.5, 5.0, 9.9, 10.0, 12.5, 15.0, 20.0 mL of HCl. Sketch the titration curve.	5
3	407.	A 10.00 ml of aliquot of 0.100 M HCl solution is titrated with 0.100 M NaOH. Calculate the pH values after adding 0.0, 2.5, 5.0, 9.9, 10.0, 12.5, 15.0, 20.0 mL of NaOH. Sketch the titration curve.	5
3	408.	A 20.00 ml of aliquot of 0.100 M NaOH solution is titrated with 0.100 M HCl. Calculate the pH values after adding 0.0, 5.0, 10.0, 15.0, 20.0, 25.0, 30.0, 40.0 mL of HCl. Sketch the titration curve.	5
3	409.	A 20.00 ml of aliquot of 0.100 M HCl solution is titrated with 0.100 M NaOH. Calculate the pH values after adding 0.0, 5.0, 10.0, 15.0, 20.0, 25.0, 30.0, 40.0 mL of NaOH. Sketch the titration curve.	5
3	410.	A 10.00 ml of aliquot of 0.100 M AcOH solution is titrated with 0.100 M NaOH. Calculate the pH values after adding 0.0, 2.5, 5.0, 9.9, 10.0, 12.5, 15.0, 20.0 mL of NaOH. Sketch the titration curve (Given K_a of AcOH= 1.75×10^{-5})	5
3	411.	A 20.00 ml of aliquot of 0.100 M AcOH solution is titrated with 0.100 M NaOH. Calculate the pH values after adding 0.0, 5.0, 10.0, 15.0, 20.0, 25.0, 30.0, 40.0 mL of NaOH. Sketch the titration curve (Given K_a of AcOH= 1.75×10^{-5})	5

3	412.	A 20.00 ml of aliquot of 0.100 M NH_4OH solution is titrated with 0.100 M HCl. Calculate the pH values after adding 0.0, 5.0, 10.0, 15.0, 20.0, 25.0, 30.0, 40.0 mL of HCl. Sketch the titration curve (Given K_b of $NH_4OH = 1.81 \times 10^{-5}$)	5
3	413.	A 10.00 ml of aliquot of 0.100 M NH_4OH solution is titrated with 0.100 M HCl. Calculate the pH values after adding 0.0, 2.5, 5.0, 7.5, 10.0, 12.5, 15.0, 20.0 mL of HCl. Sketch the titration curve (Given K_b of $NH_4OH = 1.81 \times 10^{-5}$)	5
3	414.	A 20.00 ml of aliquot of 0.100 M NaCl solution is titrated with 0.100 M silver nitrate. Calculate the pCl values after adding 0.0, 5.0, 10.0, 15.0, 20.0, 25.0, 30.0, 35.0 mL of the titrant. Sketch the titration curve (given $K_{sp} = 1.8\times 10^{-10}$)	5
3	415.	A 10.00 ml of aliquot of 0.100 M NaCl solution is titrated with 0.100 M silver nitrate. Calculate the pCl values after adding 0.0, 2.5, 5.0, 7.5, 10.0, 12.5, 15.0, 17.5, 20.0 mL of the titrant. Sketch the titration curve (given $K_{sp} = 1.8\times 10^{-10}$)	5
3	416.	Sketch the titration curve for 10.0 ml of 0.1 M Fe (II) and 0.1 M Ce(IV) in matrix of 1 M $HClO_4$ at 2.5, 5.0, 7.5, 10.0, 12.5, 15.0, 17.5, 20.0 ml of Ce (IV) (standard redox potential of Fe (III)/Fe (II) = 0.761 V and Ce (IV)/Ce (III) = 1.70 V)	5
3	417.	Sketch the titration curve for 20.0 ml of 0.1 M Fe (II) and 0.1 M Ce(IV) in matrix of 1 M $HClO_4$ at 5.0, 10.0, 15.0, 20.0, 25.0, 30.0, 35.0, 40.0 ml of Ce (IV) (standard redox potential of Fe (III)/Fe (II) = 0.761 V and Ce (IV)/Ce (III) = 1.70 V)	5
3	418.	Sketch the titration curve for 25.0 ml of 0.1 M Fe (II) and 0.1 M Ce(IV) in matrix of 1 M $HClO_4$ at 5.0, 10.0, 15.0, 20.0, 25.0, 30.0, 35.0, 40.0 ml of Ce (IV) (standard redox potential of Fe (III)/Fe (II) = 0.761 V and Ce (IV)/Ce (III) = 1.70 V)	5
3	419.	Sketch the titration curve for 30.0 ml of 0.1 M Fe (II) and 0.1 M Ce(IV) in matrix of 1 M $HClO_4$ at 7.0, 14.0, 21.0, 28.0, 30.0, 35.0, 42.0, 48.0, 54.0, 60.0 ml of Ce (IV) (standard redox potential of Fe (III)/Fe (II) = 0.761 V and Ce (IV)/Ce (III) = 1.70 V)	5
1	420.	Give the classification of analytical methods	6
1	421.	Write the classification of analytical methods	6
1	422.	An analyst reported the following percentage of FeO in a sample, 16.65, 16.70, 16.58, 16.60, 16.38 and 16.63. From this set of results calculate mean, median, average deviation, relative average deviation and coefficient variation.	6
1	423.	Explain different types and sources of errors observed in chemical analysis.	6
1	424.	Classify and describe different types and sources of errors observed in chemical analysis.	6
1	425.	Analysis of a sample of silver alloy gave the following percentage values for the silver content: 7.08, 7.21, 7.12, 7.09, 7.14, 7.07, 7.14, 7.18, and 7.11. Calculate the mean, median, standard deviation, relative standard deviation and coefficient of variance for the set of data.	6

1	426.	An analysis to determine the percentage weight of copper in an ore sample gives the following results: 16.45%, 16.30%, 16.64%, 16.67%, 16.70% and 16.49% (w/w). Clculate the mean, median, standard deviation, range coefficient of variance for this data set.	6
1	427.	What is F-test and t-test? For what purpose are these tests applied? Write the steps involved in applying them.	6
1	428.	Write a note on the steps involved in applying F-test and t-test.	6
1	429.	A new high performancwe liquid chromatographic method for the determination of Psuedoephedrine in a pharmaceutical product at two different levels compared with an established method with the following results. (i) Method 1: 59.9, 59.3, 60.4, 59.6, 60.2, 60.7 (ii) Method 2: 58.6, 58.3, 60.5, 60.7, 59.2, 59.8. Do the means of two methods differ significantly at the confidence level? (t tab = 2.228)	6
1	430.	A blood sample is sent to two laboratories to be analysed for cholesterol. The results obtained for the concentration (mg/lit) were (i) Lab 1: Mean = 243, S1 = 0.13, n1 = 11 (ii) Lab 2: Mean = 248, S2 = 0.15, n2 = 11. (1) Are the standard deviation significantly different at the 95% level? (2) Are the mean significantly different at the 95% confidence level? (Ftab = 2.98 and t tab = 2.086)	6
1	431.	What is t-test? Elaborate the steps involved in applying t-test.	6
1	432.	Write the steps involved in applying F-test and t-test.	6
1	433.	What is t-test? Indicate its significance in determining the reliability of experimental results. Outline the steps involved in applying the test.	6
1	434.	Discuss how determinate errors are propagated when arithmetic operations like addition, subtraction and multiplication, division performed.	6
1	435.	What is t-test? Outline the steps involved in applying t-test.	6
1	436.	What is Q-test? For what purpose are these tests applied? Write the steps involved in applying them.	6
1	437.	What is Q-test? For what purpose are these tests applied? Outline the steps involved in applying them.	6
1	438.	Write a note on comparison of two means and two standard deviations.	6
1	439.	Explain comparison of two means and two standard deviations.	6
1	440.	Describe comparison of two means and two standard deviations.	6

Question Paper Pattern- Model Question Paper

Q.P. Code:

St. Philomena's College (Autonomous), Mysore			
I Semester –Course M.Sc Final Examination			
Subject : Chemistry			
Title: Fundamentals of Chemical Analysis (HC)			
Time: 3 hrs			**Max Marks: 70**
PART-A			
1		**Answer any ten of the following:**	**10x2=20**
	a	Define the terms: Accuracy and Precision.	2
	b	Describe Sensitivity and Selectivity?	2
	c	An analyst determines value of 70.55% vitamin C in a fresh gooseberry that actually contains 70.25%. Calculate the absolute and relative error	2
	d	What is F-test and t-test?	2
	e	What is meant by Random sampling?	2
	f	What is organic polymeric sorbent?	2
	g	What is meant by amphiprotic solvent?	2
	h	What is the role of non-aqueous solvents in acid-base titration?	2
	i	Discuss any two applications of redox titration	2
	j	Calculate the pH of 0.1 M acetic acid (given that $K_a = 1.75 \times 10^{-5}$)	2
	k	What is ferroin indicator? Write its structure	2
	l	What is precipitation titration? Describe different types of precipitation titration with principles and indicators involved in them	2
PART-B			
		Answer any Five of the following:	**5 x 10= 50**
2	a	What is an error? Outline determinate and indeterminate errors.	5
	b	Explain briefly the classification of analytical techniques.	5
3	a	What is t-test? Elaborate the steps involved in applying t-test.	6
	b	The analysis of calcite samoleyeildingCaO percentage of 55.95, 56.00, 56.04, 56.08 and 56.23.The last value appears anomolous. Should it be rejected or retained? Q tab at 90% confidence level is 0.54	4

4	a	A new high performancwe liquid chromatographic method for the determination of Psuedoephedrine in a pharmaceutical product at two different levels compared with an established method with the following results. (i) **Method 1**: 59.9, 59.3, 60.4, 59.6, 60.2, 60.7 (ii) **Method 2**: 58.6, 58.3, 60.5, 60.7, 59.2, 59.8. Do the means of two methods differ significantly at the confidence level? (t tab = 2.228)	5
	b	Write a note on designing a sample plan.	5
5	a	Write a note on leveling effect.	5
	b	Explain the process digestion and decomposition process for bringing solid sample into solutions.	5
6	a	Discuss the titrimetric analysis in the determination of carboxylic acid, phenols and amine.	5
	b	In an analysis for the inorganic ash content of a breakfast cereal required a sample of 1.5 g to establish a relative standard deviation for sampling of ±2.0%. How many samples are needed to obtain a relative sampling error of no more than 0.80% at the 95% confidence level?	5
7	a	Explain the difference between absolute stability constant and conditional stability constant. How they are related. Add a note on the importance of conditional stability constant in EDTA titration.	5
	b	Sketch the titration curve for 50.0 ml of 0.1 M Fe (II) and 0.1 M Ce(IV) in matrix of 1 M $HClO_4$ at 10.0, 20.0, 30.0, 40.0, 50.0, 60.0, 70.0, 80.0 ml of Ce (IV) (standard redox potential of Fe (III)/Fe (II) = 0.761 V and Ce (IV)/Ce (III) = 1.70 V)	5
8	a	Propose schemes for the determination of following by acid base titrimetry a) SO_2 in air b) Nitrogen in ammonium salt c) Ester group in organic compounds d) carbonyl group in organic compounds e) alcohol group of organic compounds	5
	b	What are precipitation titrations? Write the importance of K_{sp} and application of precipitation titration for the determination of chloride content by Mohr's method	5

*****